RiverSight: public river webcams as calibrated water-level instruments

Technical report, OpenCV AI Competition 2026 (powered by AWS). Draft v2, 2026-10-24. Live: https://d2aes2r6og8aap.cloudfront.net. Sections marked *pending* fill in as the sprint lands; every number here is generated from, or traceable to, a committed file in `results/` (`python scripts/check_report_numbers.py`).

1. Problem

The USGS runs hundreds of public HIVIS webcams that look at rivers next to official stream gauges. A gauge gives one number, stage in feet, every 15 minutes. When it goes down, nobody knows the water level at that site until a technician fixes it, and that can happen in the middle of a flood. When it reads wrong, nothing independent says so. The camera beside it is a second instrument that nobody reads, because pixels are not feet.

Existing camera water-level systems read a staff plate, need a surveyed site, or run on a single camera with hand tuning (`docs/PRIOR_ART.md`). None of them scale to a fleet of cameras that were never set up for measurement.

RiverSight reads stage from these cameras with no plate and no survey, by learning from the gauge's own history which parts of each scene go under water at which stage. It also checks itself: it notices when a camera has moved, when the lens is fogged, and when the camera and gauge disagree, and it hands those cases to an agent that a human supervises.

2. Users and decisions

Three people, each making a real decision with a camera they already have but cannot read. For each: what they see, the decision it supports, and what it does not replace.

County emergency manager (a flood-prone Texas county, say, with a USGS camera on a creek crossing).

- *Sees:* the site page with the latest frame and the wet area drawn on it; a status pill (OK, provisional, above range, night); a slider showing what is under water at any stage; the NWS forecast crest drawn into today's picture where a forecast exists; a drafted alert waiting for approval when the water rises.
- *Decision:* whether to close a low-water crossing or send a crew to look, especially when the gauge has gone silent or is hours old.
- *Does not replace:* NWS warnings or the county's own gauges and spotters. Alerts are drafted, never sent without a person. Above the calibrated range the camera says "at least X ft", not a number.

NWS hydrologist or forecaster.

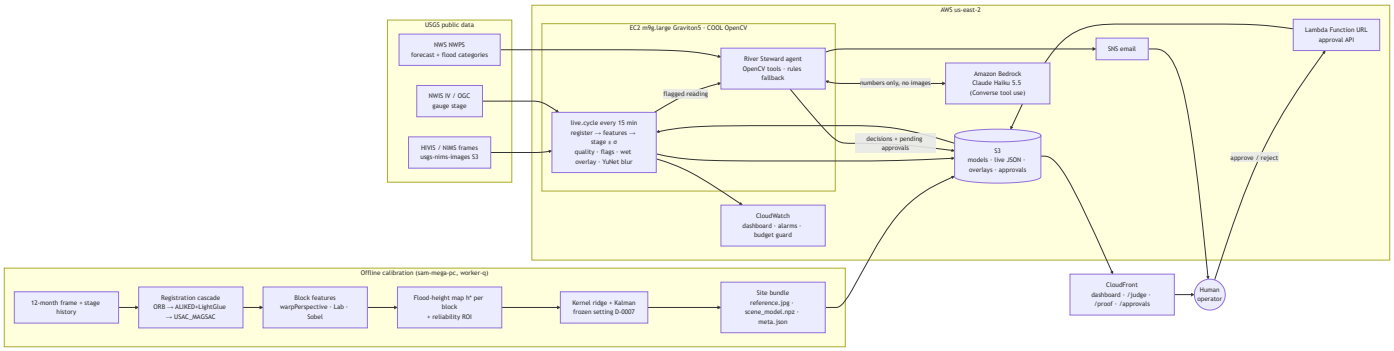
- *Sees:* flood state per camera (above or below the site's 90th percentile), with σ; which gauges are stale or disagree with their camera; the fleet triage list ranked by what needs a human.
- *Decision:* whether a gauge reading feeding a forecast is trustworthy right now, and whether a site without a working gauge is rising.
- *Does not replace:* the gauge record, the rating curve or the forecast model. A camera reading is a provisional, independent check.

USGS hydrographer at a water science center.

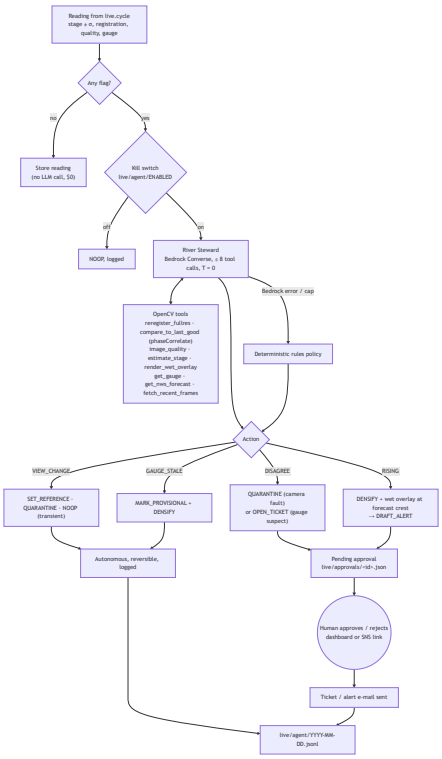
- *Sees:* cameras flagged for a view change, quarantined or re-anchored, with the registration evidence; tickets saying "gauge suspect" when a confident camera disagrees with a flat-lining or jumping gauge; provisional gap-fill during an outage, with σ.

- *Decision:* which site to visit first; whether a gauge needs a check before its data is approved; which camera needs a new reference after a re-aim.
- *Does not replace:* field measurements or the approval of official records. Gap-filled values are labelled provisional and never written into the USGS record.

3. Architecture



Offline, each site is calibrated once from up to 12 months of frames and gauge readings. Live, one Graviton worker reads the newest frame of each of the 20 calibrated cameras every 15 minutes, produces a stage ± σ plus quality flags, and writes it to S3 behind CloudFront. Flagged readings go to the River Steward agent. Details: `docs/ARCHITECTURE.md`.



4. OpenCV 5 implementation

Full table, with what breaks without each call: `docs/OPENCV5_USAGE.md`.

Registration. Every frame is registered to the site's reference: ORB + BFMatcher first, then, for the frames ORB cannot register, OpenCV 5.0's learned **ALIKED** features with the **LightGlueMatcher** (both new in 5.0, running on the new DNN engine), and a **USAC_MAGSAC** homography. The inlier ratio and reprojection error become the registration score that drives view-change detection.

Scene model (D-0006; setting frozen in D-0007).

1. `warpPerspective` puts the frame in reference geometry; `resize(INTER_AREA)` brings it to 320 × 180.
2. The frame is cut into 3,600 blocks of 4 px. Each block gets five features: Lab lightness (normalised per frame), a, b, `Sobel` gradient magnitude and lightness spread.
3. **Flood-height map.** For each block, a threshold h* on gauge stage is fitted so the block's features separate "stage below h*" from "stage above h*"; its accuracy is the block's reliability. Reliable blocks (≥ 0.80) form the region of interest. h* is also what the dashboard and the agent draw: every block with h* ≤ a given stage is under water at that stage.
4. A linear-kernel ridge regression (λ = 1) on the standardised ROI features gives a raw stage; a causal 1-D Kalman filter smooths it, and σ comes from out-of-fold residuals.

QA and privacy. Laplacian variance, glare and a dark-channel fog score flag bad frames; YuNet (`FaceDetectorYN`) blurs faces before any frame is published. `cv2.Animation` / `imwriteanimation` write the event GIFs.

Connected inundation (display). The slider and the wet overlay should show water only where water can be. A block is drawn as wet at stage s only if h* ≤ s, it is reliable, **and** it is 4-connected to the river through other such blocks. The seed is the lowest decile of h* inside the river corridor: the reliable component with the largest area × (centroid height)^2, because water lies low in a river camera's view while stray reliable specks on foliage or sky are small and high. The fill uses `cv2.connectedComponentsWithStats` on the 80 × 45 block grid (`live/bundle.py: connected_inundation`). It is display only: the stage estimate and every reported number use the unconnected model.

Arm64. The live path uses `INTER_AREA` only and pads with `copyMakeBorder` instead of `warpPerspective(BORDER_REPLICATE)`, avoiding two KleidiCV issues in the 5.0.0.93 arm64 wheels (#29794, #29816).

5. AWS deployment

Live at https://d2aes2r6og8aap.cloudfront.net since 2026-10-23, in us-east-2 (`docs/DEPLOY.md`, `infra/`).

- **Worker:** one EC2 m9g.large (Graviton5) from the **COOL G5 AMI**. A systemd timer runs one cycle over all calibrated cameras every 15 minutes. `/proof` publishes the instance-identity document, AMI id, `cv2.__file__`, version and `getBuildInformation()` hash, so a judge can check which OpenCV produced the readings.
- **Storage and front door:** a private S3 bucket (models, live readings, overlays, approvals) behind CloudFront with origin access control; the dashboard is static.
- **Agent:** Amazon Bedrock (Claude Haiku 5.5 through the `us.` inference profile, Converse tool use) for flagged readings only; a deterministic rules policy as fallback.
- **Human approval:** a Lambda Function URL behind a token; SNS e-mail to the operator.
- **Guardrails:** an AWS Budget on the $100 cap (D-0004) whose 90 % action stops every instance tagged for the project, created before the first launch; IMDSv2; no SSH (SSM only); least-privilege roles.
- **Operations:** `docs/RUNBOOK.md` (worker down, USGS down, Bedrock throttled, tear-down).

6. Evaluation

6.1 Protocol

Pre-registered before any test frame was read (D-0006):

- 20 cameras chosen for gauge history and view variety (`data/sites/kill_test.yaml`).
- Calibration: all frames before 2026-10-03 00Z (history up to 12 months, minimum 90 days). Tuning only by forward-chaining CV inside the calibration data: 36 settings, selection rule committed before any CV result existed (D-0007).
- The chosen setting and every scoring module were hashed (`results/constants.sha256`) before the test days were read.

- Test: daylight frames, 2026-10-04 00Z to 2026-10-14 00Z, scored **once** (worker-q job 2986, `scripts/scene_score.py`, which refuses to run a second time).
- Pass criteria: **P1** ≥ 10/20 sites with MAE ≤ 0.25 ft or nMAE ≤ 10 % of the calibration p5–p95 stage range; **P2** the camera beats a 24-hour-stale gauge on event frames at ≥ 70 % of sites with events; **P3** registration ≥ 90 %.

6.2 Headline results (verdict: PASS)

metric	value	source
P1 accuracy (MAE ≤ 0.25 ft or nMAE ≤ 10 %)	**14/20** sites (bootstrap 95 % interval 12–16)	`kill_test_v2_verdict.json`, `kill_test_v2_ci_summary.json`
median MAE across sites	**0.182 ft** (95 % CI 0.148–0.22)	`kill_test_v2_ci_summary.json`
median nMAE / Spearman ρ	0.092 / 0.66	`kill_test_v2_summary.json`
P2 value on event frames vs 24 h-stale gauge	**15/16** event sites	`kill_test_v2_verdict.json`
P3 registration, 30-day daylight frames	**95.7 %** (8611 of 9000)	`kill_test/registration_summary.csv`
flood-state balanced accuracy, median over 9 sites with flood frames	**0.93** vs 0.75 (24 h-stale gauge)	`kill_test_v2_summary.json`
coverage: daylight hours with a reading, median site	94.1 % (47.5 % of all hours)	`coverage_v2.csv`
fleet camera-hours of stage per day	219.4	`kill_test_v2_ci_summary.json`

CIs are a block bootstrap over test days (`report/bootstrap_coverage.py`). "Covered" means at least one frame in that hour produced a reading.

6.3 Per site

site	tier	frames	MAE ft [95% CI]	RMSE ft	nMAE	P1	event MAE: camera / 24 h LKV	daylight hours covered
Cowhouse Ck at Old Georgetown Rd nr Ft Hood (TX)	C	236	0.212 [0.141–0.320]	0.288	0.15	✓	0.64 / 1.01	100.0 %
Tenmile Ck at Parkinson Rd nr Ferris (TX)	C	119	0.617 [0.539–0.679]	0.685	0.33	✗	0.20 / 0.84	90.8 %
Buffalo Bayou at State HWY 6 near Addicks (TX)	B	113	0.268 [0.182–0.364]	0.353	0.09	✓	0.23 / 1.06	85.8 %
Clear Fk Trinity Rv at Ft Worth (TX)	B	120	0.133 [0.098–0.179]	0.171	0.11	✓	0.20 / 0.91	91.7 %
White Rk Ck at Greenville Ave Dallas (TX)	C	58	0.782 [0.695–0.880]	0.833	0.72	✗	0.56 / 0.52	40.8 %
Langham Creek at Addicks Reservoir Outflow near Addicks (TX)	B	211	0.487 [0.384–0.569]	0.661	0.10	✗	0.57 / 2.02	95.0 %
East Branch Pecatonica River near Blanchardville Bullet (WI)	A	113	0.091 [0.071–0.115]	0.108	0.02	✓	0.07 / 0.34	94.2 %
Swan Creek at Lalor Road near Fitchburg (WI)	C	113	0.060 [0.046–0.076]	0.073	0.04	✓	—	94.1 %
Root River near Pilot Mound (MN)	C	94	0.451 [0.362–0.541]	0.482	0.10	✗	—	78.3 %
East Branch Delaware River at Hancock Staff (NY)	A	114	0.065 [0.049–0.081]	0.087	0.04	✓	0.07 / 0.59	95.0 %
Neversink River at Godeffroy (NY)	B	114	0.060 [0.036–0.095]	0.086	0.04	✓	0.11 / 0.51	95.0 %
Stony Brook at Princeton (NJ)	B	111	0.106 [0.074–0.145]	0.138	0.06	✓	0.13 / 1.09	92.5 %

Big Brook near Marlboro (NJ)	C	58	0.146 [0.103–0.200]	0.173	0.17	✓	0.11 / 0.40	48.3 %
East Fork Little Miami River at Perintown (OH)	B	120	0.475 [0.359–0.588]	0.545	0.11	✗	—	94.8 %
Bean Creek at Powers (OH)	B	113	0.270 [0.204–0.339]	0.312	0.05	✓	0.22 / 0.58	89.6 %
Animas River at Durango (CO)	B	230	0.152 [0.113–0.185]	0.167	0.07	✓	0.02 / 0.29	100.0 %
Sand Creek at Mouth nr Commerce City (CO)	C	237	0.069 [0.060–0.078]	0.080	0.09	✓	—	100.0 %
Salt Creek at Wood Dale (IL)	B	119	0.091 [0.065–0.118]	0.121	0.02	✓	0.14 / 0.90	99.1 %
Brandywine Creek at Chadds Ford (PA)	C	233	0.308 [0.210–0.405]	0.387	0.19	✗	0.37 / 1.92	99.2 %
First Mallard Branch nr Fairfield (CA)	B	208	0.391 [0.308–0.478]	0.523	0.09	✓	0.35 / 0.58	90.8 %

Per-site time series and flood-height maps: `results/kill_test_v2/`. Tiers follow pre-registered rules (D-0007): A = sites with a staff plate or structure in view, C = poor test-day registration, weak CV correlation or under 90 calibration days.

6.4 Against baselines: where the camera adds value

comparison (all scored frames unless noted)	camera better at
constant (calibration median)	16/20
gauge 24 h stale (persistence)	10/20
gauge 15 min stale (persistence)	0/20
event frames (|Δ24h| ≥ 0.25 ft): gauge 24 h stale, P2	15/16 event sites
gauge outage of 6 h: last-known value	3/20
gauge outage of 24 h: last-known value	10/20
gauge outage of 72 h: last-known value	15/20
gauge outage of 7 days: last-known value	18/20

The camera never beats a 15-minute-old gauge reading. Its value appears when the gauge is down or stale: after a 72-hour outage it beats the last known value at most sites, and after 7 days at nearly all. That is the product claim: **gap-fill during gauge outages, flood state, and camera QA**, not live replacement.

6.5 Ablations and harder settings

Registration, per site. ORB alone fails on most cameras; ALIKED + LightGlue carries the cascade. Over the 12-month history, a bank of per-view references (one per run of frames the main reference could not register) adds little at fleet scale: it matters at the two cameras that were re-aimed most (Pecatonica, Sand Creek), which is why re-anchoring is an agent action and not a default (`report/registration_ablation.py`).

site	30 d: ORB	+ ALIKED+LightGlue	+ chaining	12 mo: main reference	+ reference bank	usable for calibration
Cowhouse Ck at Old Georgetown Rd nr Ft Hood (TX)	43.3 %	99.7 %	99.7 %	97.2 %	97.2 %	97.2 %
Tenmile Ck at Parkinson Rd nr Ferris (TX)	45.7 %	100.0 %	100.0 %	99.1 %	99.1 %	99.1 %
Buffalo Bayou at State HWY 6 near Addicks (TX)	28.0 %	98.9 %	98.9 %	95.2 %	95.2 %	95.2 %
Clear Fk Trinity Rv at Ft Worth (TX)	35.3 %	99.2 %	99.2 %	98.4 %	98.4 %	98.4 %
White Rk Ck at Greenville Ave Dallas (TX)	19.4 %	72.9 %	79.5 %	49.8 %	49.8 %	42.5 %

Langham Creek at Addicks Reservoir Outflow near Addicks (TX)	10.5 %	99.0 %	99.0 %	99.2 %	99.2 %	99.2 %
East Branch Pecatonica River near Blanchardville Bullet (WI)	4.5 %	85.5 %	85.5 %	61.6 %	75.5 %	61.8 %
Swan Creek at Lalor Road near Fitchburg (WI)	36.4 %	93.8 %	93.8 %	96.8 %	96.8 %	96.8 %
Root River near Pilot Mound (MN)	19.6 %	89.9 %	91.8 %	75.6 %	75.6 %	73.9 %
East Branch Delaware River at Hancock Staff (NY)	100.0 %	100.0 %	100.0 %	99.8 %	99.8 %	99.8 %
Neversink River at Godeffroy (NY)	78.7 %	98.1 %	98.1 %	97.0 %	97.0 %	97.0 %
Stony Brook at Princeton (NJ)	40.3 %	92.2 %	92.2 %	94.0 %	94.0 %	94.0 %
Big Brook near Marlboro (NJ)	3.5 %	55.6 %	71.3 %	22.3 %	22.3 %	20.8 %
East Fork Little Miami River at Perintown (OH)	89.7 %	100.0 %	100.0 %	99.4 %	99.4 %	99.4 %
Bean Creek at Powers (OH)	62.6 %	93.3 %	93.6 %	94.1 %	94.1 %	94.1 %
Animas River at Durango (CO)	11.6 %	98.5 %	99.7 %	99.5 %	99.5 %	99.5 %
Sand Creek at Mouth nr Commerce City (CO)	10.1 %	92.6 %	99.9 %	26.3 %	32.8 %	26.3 %
Salt Creek at Wood Dale (IL)	41.6 %	98.9 %	98.9 %	47.4 %	47.4 %	46.7 %
Brandywine Creek at Chadds Ford (PA)	99.6 %	99.7 %	99.7 %	92.6 %	92.6 %	92.6 %
First Mallard Branch nr Fairfield (CA)	26.7 %	93.0 %	93.0 %	83.9 %	85.8 %	80.9 %
fleet (9000 / 38668 daylight frames)	**39.6 %**	**94.0 %**	**95.7 %**	**80.5 %**	**81.4 %**	**79.6 %**

ablation	result	source
registration: ORB + MAGSAC alone vs full cascade	39.6 % vs 95.7 % of 9000 daylight frames; ALIKED + LightGlue recovered 4892 frames	`kill_test/registration_summary.csv`
estimator: classical transect waterline vs scene model, 20-day fit-window CV, sites ≤ 0.25 ft	1/20 vs 9/19 (scene median 0.278 ft)	`kill_test/fit_window_cv.csv`, `scene/gate_fit20.csv`
calibration horizon: year-long forward-chaining CV (frozen setting)	P1 at 9/20, median MAE 0.38 ft, median nMAE 0.145; 36 settings tried	`scene/tune_cv_summary.csv`

The year-long CV is the honest long-run number. Its error is about twice the test-day error. Two causes are confounded: the scene changes through the year (vegetation, sun angle, sediment), and the early folds train on only one or two months (`results/failures/seasonal_monthly_mae.csv`). Periodic recalibration is planned for the live models (they are refit on all data up to export, and never mixed into the frozen numbers).

6.6 Head-to-head at Hancock, NY

StageSight, another entry, reads the staff plate on the same USGS camera (Hancock NY, 01421500) and reports two floods. The two systems make different assumptions: StageSight needs a red-marked plate in view and no gauge history; RiverSight needs gauge history and no plate. To compare fairly we refit our scene model with the frozen setting on all calibration frames **except ±7 days around each event** (leave-event-out), and score the event's registered daylight frames against the gauge (`report/headtohead_hancock.py`, `results/headtohead_hancock/`). Our windows are the UTC days of each event; StageSight's exact frame sets may differ.

	29–30 Jul 2026 flood	4–5 Sep 2026
gauge range in window, ft	9.56–18.08	10.87–11.97
our calibration max after leaving the event out, ft	11.97	18.08
RiverSight RMSE, ft [95 % CI]	4.32 [3.44–5.07]	0.46 [0.36–0.54]
RiverSight MAE / within 0.5 ft	3.29 / 21.4 %	0.39 / 61.5 %
RiverSight flood-state recall (camera above calibration p90 when the gauge is)	1.00	0.92
baseline: gauge value at window start held (outage at event start), RMSE	5.53	0.41
baseline: calibration median, RMSE	5.40	1.76

RiverSight coverage: hours of the window / listed daylight frames with a reading	58.3 % / 93.3 %	54.2 % / 100.0 %
StageSight RMSE as reported, ft (not reproduced by us)	0.26 [0.20–0.31]	0.34
StageSight coverage as reported	2.4 % of hourly frames (whole archive)	

Reading this honestly.

- On the moderate September event we are worse than StageSight's reported figure, and slightly worse than simply holding the gauge's value from the start of the event.
- On the July flood we fail: with that flood left out, the calibration never saw a stage above 11.97 ft, and the scene model cannot extrapolate. It saturates near its calibrated ceiling while the river rises above it. Its Kalman σ stays small, so it is confidently wrong.
- What survives is flood state: the camera put itself above the calibration p90 on every July frame where the gauge was there.
- A plate reader measures stages it has never seen; a model calibrated from history does not. This is the main limitation of the approach (§7). The live system now has to flag readings at the edge of the calibrated range instead of reporting them as numbers.
- Future work: every flood the gauge records could extend the camera's calibrated range, because the site bundle can be refit on new data. Today that refit is a manual export (`scripts/export_site_models.py`), not a scheduled job.
- Our advantage is elsewhere: 20 cameras without plates, and coverage of most daylight hours, against 2.4 % of hourly frames reported for the plate reader.

6.7 Agent evaluation (River Steward)

W2's frozen evaluation (`results/agent_eval/REPORT.md`, `LLM_ANALYSIS.md`): every arm runs every episode; task success = the right primary action without forbidden ones. Claude models through Amazon Bedrock (the 5.5 models were not yet available to the project, so the evaluation used Claude Sonnet 4.6 and Haiku 4.5).

arm	model	repeats	task success (min–max)	real / synthetic	false actions on clean controls	fleet triage top-1	p50 latency s	est. $ total
deterministic rules	—	3	93.9 % (93.9–93.9)	98.4 % / 85.7 %	0.0 %	91.7 %	1.3	0.0
LLM agent + OpenCV tools	Claude Sonnet 4.6	1	71.4 % (71.4–71.4)	79.4 % / 57.1 %	3.8 %	100.0 %	55.7	5.128
LLM agent + OpenCV tools	Claude Haiku 4.5	3	58.8 % (56.1–61.2)	54.5 % / 66.7 %	7.7 %	44.4 %	42.0	4.326
LLM agent, flags only (no OpenCV evidence)	Claude Haiku 4.5	3	51.4 % (49.0–55.1)	47.6 % / 58.1 %	15.4 %	52.8 %	41.2	4.126

Episodes: 98 single-camera (63 real, 35 synthetic and labelled, 26 clean controls) + 12 fleet cycles; expected actions written before any arm ran; set frozen with sha256 (`results/agent_eval/FROZEN.json`). OpenCV-evidence effect (same LLM with vs without OpenCV tools, paired): +7.5 points (95 % CI -1.4 to 16.3).

Reading this honestly.

- The deterministic rules policy beats both LLM agents on this episode set. The rules were written by the same people who wrote the expected actions, so this partly measures agreement with the authors, but it is still the best policy we have: the live system decides with rules first.
- The OpenCV evidence clearly matters where the decision depends on what the image shows. With the tools, the LLM re-anchors to a banked reference, recognises night frames and keeps muddy-water frames during a stale gauge. The overall effect is positive, but its confidence interval includes zero, and on gauge flatlines the evidence misleads the LLM.
- The LLMs' main failure is under-escalation: missing `DRAFT_ALERT` on real rises. Out-of-range readings were the other common miss. The LLM is therefore used as a second opinion on hard cases, never as the only decider (Q-0011).

- Flood replays through the live cycle drafted an alert in the first cycle above the site's p90 (`results/agent_eval/replay/`).
- Limits (from W2): gauge-stale episodes are synthetic dropouts on real frames; re-anchoring is tested mostly on one camera (Pecatonica); costs are list-price estimates.

6.8 COOL on Graviton

COOL G5 vs stock pip OpenCV 5.0.0.93 on the same m9g.large, same Python and numpy, the live per-frame path stage by stage plus the fleet cycle (`results/cool_bench/cool_g5_m9g_2026-10-24/`; full tables and method: `docs/COOL_BENCHMARK.md`).

	COOL vs stock
fleet cycle, 880 frames, 2 cores	1.0067× (1.4029 vs 1.3936 frames/s)
ORB + MAGSAC registration, 1 / 2 threads	1.16× / 1.21×
whole per-frame path, p50, 1 / 2 threads	1.13× / 1.21×
resize, cvtColor, Sobel, warpPerspective, imdecode	0.975× – 1.011×
ALIKED + LightGlue (≈ 98 % of stage time)	0.996× / 0.999×
outputs	bit-identical pixels; stage estimates within 0.00044 ft
$ per camera-month at 15 min, incl. COOL fee	$0.0586 vs $0.0562
one m9g.large covers	~880 cameras at 15 min (627 s per cycle)

Honest reading: end to end, COOL is flat on this workload, because the cycle is dominated by the learned matcher, which COOL does not speed up. The stock aarch64 wheel already ships KleidiCV, and our images are small. COOL's real gain is in ORB registration, which most frames go through. Because the outputs are identical and the cost difference is a fraction of a cent per camera-month, the live worker runs COOL. The whole gauged HIVIS fleet fits on one Graviton instance either way.

6.9 Prospective live evaluation

Pre-registered in D-0010 §1 before the first COOL cycle: the live bundles as first deployed (`results/live_prospective/frozen_models.json`) are scored on every daylight reading with registration OK, against the USGS reading closest in time, until the code freeze. Saturated (`OUT_OF_RANGE`) readings are counted, not scored. Baseline: the gauge's value 24 hours earlier. Forward in time, out of sample, reported whatever it shows; refreshed from `results/live_prospective/` (W1, `scripts/score_prospective.py`).

Window: 2026-10-23 21:28 UTC (first COOL cycle) to 2026-10-24 02:30 UTC so far; submission cut 2026-10-26 17:00 UTC. Models: frozen live models (results/live_prospective/frozen_models.json). Match: closest USGS IV reading within ±20 min of the frame. The camera beats a 24-hour-old gauge value at 0 of 13 scored sites so far. That matches the frozen test: in quiet conditions persistence is hard to beat; the camera's value is outages, flood state and QA.

	readings	**daylight**	**scored**	**saturated (gauge trusted)**	**coverage**	**camera MAE ft**	**median AE ft**	**within ±0.25 ft**	**24 h last-known MAE ft**
fleet	146	66	51	2	77.3 %	0.274	0.221	51.0 %	0.077
First Mallard Branch nr Fairfield (CA)	18	17	15	2	88.2 %	0.320	0.339	20.0 %	0.179
Sand Creek at Mouth nr Commerce City (CO)	14	12	12	0	100.0 %	0.154	0.174	100.0 %	0.023
Cowhouse Ck at Old Georgetown Rd nr Ft Hood (TX)	17	7	7	0	100.0 %	0.119	0.124	100.0 %	0.006
Langham Creek at Addicks Reservoir Outflow near Addicks (TX)	17	5	5	0	100.0 %	0.566	0.507	0.0 %	0.068
Salt Creek at Wood Dale (IL)	5	2	2	0	100.0 %	0.231	0.231	50.0 %	0.070

Brandywine Creek at Chadds Ford (PA)	17	2	2	0	100.0 %	0.447	0.447	0.0 %	0.045
Root River near Pilot Mound (MN)	5	2	2	0	100.0 %	0.413	0.413	0.0 %	0.045
East Branch Delaware River at Hancock Staff (NY)	4	1	1	0	100.0 %	0.057	0.057	100.0 %	0.040
Buffalo Bayou at State HWY 6 near Addicks (TX)	4	1	1	0	100.0 %	0.296	0.296	0.0 %	0.080
Clear Fk Trinity Rv at Ft Worth (TX)	4	1	1	0	100.0 %	0.540	0.540	0.0 %	0.080
Tenmile Ck at Parkinson Rd nr Ferris (TX)	4	1	1	0	100.0 %	0.044	0.044	100.0 %	0.010
White Rk Ck at Greenville Ave Dallas (TX)	4	1	1	0	100.0 %	0.462	0.462	0.0 %	0.040
Swan Creek at Lalor Road near Fitchburg (WI)	4	1	1	0	100.0 %	0.078	0.078	100.0 %	0.010

Flags raised: VIEW_CHANGE 13, LOW_QUALITY 8, GAUGE_STALE 0, DISAGREE 14, RISING 23, OUT_OF_RANGE 16. Agent actions: DENSIFY 23, QUARANTINE 13, OPEN_TICKET 13, NOOP 8, MARK_PROVISIONAL 6.

6.10 Onboarding new cameras and real gauge outages

How much gauge history does a new camera need? For 10 kill-test cameras, the frozen setting was refit on only the last N days of history before the same two test months (W5, `results/onboarding/fewshot*.csv`; history before 2026-10-03 only, so the frozen test days are untouched).

days of calibration	14	30	60	90	180	365
median MAE ft	0.238	0.236	0.223	0.254	0.227	0.249
median nMAE	0.113	0.089	0.102	0.131	0.137	0.150
P1 sites (of 10)	7	9	9	7	7	6

About a month of overlap with the gauge is already enough; more history does not help the frozen setting, and a full year is slightly worse, consistent with the drift in §6.12. This makes onboarding the rest of the gauged HIVIS fleet cheap. W5's pipeline (`scripts/onboard_*.py`) runs the same frozen-setting code end to end.

Real gauge outages (W5, `results/real_outages/README.md`). Over the last 12 months, 531 of the 875 gauges behind gauged HIVIS cameras had at least one outage of 6 h or more: 3,734 outages, with a median length of 15.0 h, 984 lasting a day or more and 303 a week or more. For 795 of the outages checked, the camera kept posting daylight frames while the gauge was dark. Gauges do go dark, and the cameras usually keep running.

At our 20 calibrated sites, few outages are scorable, and in most of them the river barely moved. For hard gaps scored when the gauge came back (refit without the outage ± 7 days), the camera was closer than the last known value in 1 of 16. In 14 of those the river moved less than 0.25 ft, and holding the last reading is nearly exact. Where the river did move: at Clear Fork Trinity (56 h gap) the camera bridged it, 0.11 ft against 0.34 ft for the last known value; at Root River (758 h, winter ice) it failed, 1.34 against 0.35 ft. So far that is one real outage bridged and one failure, not a statistic. The frozen-test gap-fill result (§6.4) remains the main evidence, and the outage catalogue shows how often the situation arises.

6.11 Model v2 candidate: DINOv2 features in `cv2.dnn`

W4 (`results/v2/README.md`). v2 adds a second view of each registered frame: DINOv2-S/14 patch tokens computed by OpenCV 5's DNN engine (`cv2.dnn.readNetFromONNX`; checked against onnxruntime). The tokens go through a fixed, label-free PCA fitted on cameras outside the evaluation, then get the same flood-height-map reliability selection and ridge, combined with v1 at equal weight. Every other knob is the frozen D-0007 setting. It was evaluated with exactly the D-0007 protocol

(monthly forward-chaining CV inside the calibration history; v1 reproduces the frozen numbers exactly), and the selection rule was committed before any v2 result existed.

year-long forward-chaining CV, 20 sites	v1 (D-0007)	v2 selected (`v1+dino_roi@0.5+sun`)
median MAE ft	0.380	**0.252** (-33.7 %)
mean MAE ft	0.389	0.304
median nMAE	0.145	0.122
sites meeting P1	9/20	11/20
sites improved	—	19/20 (worst site +0.015 ft)

v2 is better in every calendar month, most of all in the foliage months that drive v1's drift. The ablations show the DINOv2 features do the work; the seasonal tweaks (sun-position correction, recency weighting) add nothing or hurt. Limits: this is still CV inside one year, the frozen kill-test numbers are not recomputed, and the forward-in-time check is v2's shadow scoring on the live stream (D-0010 §2). Whether v2 is served is decided on Oct 25 (Cowork).

6.12 Failure cases

Nine concrete cases, each with an image, its numbers and a paragraph: `results/failures/README.md`. Night/IR frames; a hazy lens on a high-flow day (the QA metric cannot separate lens haze from turbid water); a scene changed by a flood (White Rock); a re-aimed camera (Pecatonica); a model that learned the sun instead of the water (Tenmile); a far bank with almost no signal (Root River); tides (First Mallard); a flood above the calibrated range (Hancock); and month-to-month drift.

7. Limitations (not claimed)

- **Not a replacement for gauges or official warnings.** The camera beats 15-minute persistence at no site.
- **Daylight only.** Night and IR frames are not scored, so the camera covers about half of all hours.
- **Needs gauge history.** Calibration uses the co-located gauge, so it does not transfer to ungauged cameras yet.
- **No extrapolation above the calibrated range.** A flood higher than anything in the calibration data is read as roughly the calibrated maximum, with an overconfident σ (§6.6). Flood state still fires.
- **Seasonal drift.** See §6.5.
- **Cameras move.** Several cameras have history periods taken from a different view (`results/kill_test_v2_summary.json`). The live system detects view changes and the agent re-anchors or quarantines.
- **Quiet test windows.** Several P1 passes come from flat test windows where simple baselines are better still (Swan Creek, Sand Creek, Big Brook; Cowhouse vs 24 h persistence).
- **Failures:** Tenmile ($\rho < 0$), White Rock (scene changed after a flood), Langham, Root River, East Fork Little Miami and Brandywine miss P1.
- **Synthetic faults** in the agent evaluation are labelled as such.

8. Responsible use

Draft; full version in `docs/RESPONSIBLE_USE.md`.

- Readings are labelled provisional; alerts and tickets are drafted by the agent and sent only after a human approves.
- No images go to the LLM, only numbers and short summaries. Faces are blurred (YuNet) before any frame is published.
- USGS imagery is public domain. We do not redistribute raw frames; derived overlays carry a credit line.
- Spending is capped by an AWS Budget with an automatic stop.

Reproduce

`docs/REPRODUCE.md` (*pending*): fresh clone → tests → summary and CIs from committed predictions → COOL bench → deploy.